

1. [Theory] Which statement best describes an `if-elif-else` chain?
 - A) Every true condition is executed
 - B) Only the first true branch is executed
 - C) Only the last true branch is executed
 - D) All branches are checked and executed
2. [Theory] What is the main purpose of indentation in Python conditional and loop blocks?
 - A) It improves only readability
 - B) It defines the block of statements controlled by the statement
 - C) It converts code into machine language
 - D) It is optional
3. [Interview] What does `break` do inside nested loops?
 - A) Stops all loops
 - B) Stops the nearest enclosing loop
 - C) Skips only the current statement
 - D) Restarts the outer loop
4. [Interview] What does `continue` do in a loop?
 - A) Terminates the program
 - B) Terminates the loop
 - C) Skips the remaining statements of the current iteration and proceeds to the next iteration
 - D) Repeats the current iteration forever
5. [Theory] What is the difference between `=` and `==` in Python?
 - A) Both perform comparison
 - B) `=` assigns a value, while `==` compares values
 - C) `==` assigns and `=` compares
 - D) Both perform assignment
6. [Interview] Which condition is true for an even positive integer `n` greater than 2?
 - A) $n \% 2 == 1$
 - B) $n \% 2 == 0$ and $n > 2$
 - C) $n // 2 == 0$
 - D) $n \% 2 != 0$
7. [Theory] What values are produced by `range(2, 10, 3)`?
 - A) 2, 3, 4, 5, 6, 7, 8, 9
 - B) 2, 5, 8
 - C) 2, 5, 8, 10
 - D) 3, 6, 9
8. [Interview] Which situation can cause a `while` loop to become infinite?
 - A) Its condition eventually becomes False
 - B) The loop variable is updated toward termination

C) The condition remains True because the relevant state never changes

D) The loop contains an `if` statement

9. [Theory] In Python, what is the result of `not (True and False)`?

A) True

B) False

C) None

D) Error

10. [Interview] What is the advantage of using `elif` instead of several independent `if` statements when only one category should be selected?

A) It guarantees every condition executes

B) It prevents later branches from being selected after a previous condition succeeds

C) It removes indentation

D) It makes loops unnecessary

11. [Theory] What does `//` perform in Python?

A) Normal division

B) Modulo

C) Floor division

D) Exponentiation

12. [Interview] If `break` occurs inside an inner loop, what normally happens to the outer loop?

A) The outer loop is also immediately terminated

B) The outer loop continues normally

C) The program always terminates

D) The outer loop becomes infinite

13. [Theory] Which expression checks whether `n` is divisible by both 3 and 5?

A) $n \% 3 == 0$ or $n \% 5 == 0$

B) $n / 3 == 0$ and $n / 5 == 0$

C) $n \% 3 == 0$ and $n \% 5 == 0$

D) $n // 3 == 0$ and $n // 5 == 0$

14. [Interview] Why is `if x > 10` different from `if x >= 10`?

A) The first includes 10 and the second excludes it

B) The first excludes 10 and the second includes it

C) They are always identical

D) Both are syntax errors

15. [Theory] What happens when a `for` loop completes normally without encountering `break` and has an `else` clause?

A) The `else` clause executes

B) The `else` clause is skipped

C) The loop restarts

D) Python raises an error

16. [Interview] What is a common logical mistake when writing a range check for `10 <= x <= 20`?

- A) Using two comparisons
- B) Accidentally using `or` when both boundaries must be satisfied
- C) Using indentation
- D) Using `elif`

17. [Theory] What does `range(5, 0, -1)` generate?

- A) 5, 4, 3, 2, 1
- B) 5, 4, 3, 2, 1, 0
- C) 0, 1, 2, 3, 4
- D) 4, 3, 2, 1, 0

18. [Interview] In a nested `if`, why can the indentation level change the program's logic?

- A) Indentation only changes comments
- B) It determines which statements belong to which conditional block
- C) It changes variable types
- D) It changes arithmetic precedence

19. [Theory] Which operator has higher logical precedence in Python?

- A) `or` has higher precedence than `and`
- B) `and` has higher precedence than `or`
- C) `not` has lower precedence than `or`
- D) All have identical precedence

20. [Interview] What is the best way to trace a difficult loop during an interview?

- A) Guess the output
- B) Track the loop variable and important variables after every iteration
- C) Ignore conditional branches
- D) Count only the first iteration

21. [Output Logic] What is printed?

```
x = 8
if x > 5:
    if x % 2 == 0:
        print("A")
    else:
        print("B")
else:
    print("C")
```

- A) A
- B) B
- C) C
- D) Nothing

22. [Output Logic] What is printed?

```
n = 5
while n > 1:
```

```
if n % 2 == 0:
    n -= 2
else:
    n -= 1
print(n)
```

- A) 0
- B) 1
- C) 2
- D) 3

23. [Output Logic] What is printed?

```
for i in range(1, 5):
    if i == 3:
        continue
    print(i, end=" ")
```

- A) 1 2 3 4
- B) 1 2 4
- C) 2 4
- D) 1 3

24. [Output Logic] What is printed?

```
x = 12
if x % 2 == 0:
    x //= 2
elif x % 3 == 0:
    x += 3
else:
    x -= 1
print(x)
```

- A) 5
- B) 6
- C) 9
- D) 11

25. [Interview Logic] A program must print `Pass` only when marks are at least 40 AND attendance is at least 75. Which condition is correct?

- A) marks >= 40 or attendance >= 75
- B) marks > 40 and attendance > 75
- C) marks >= 40 and attendance >= 75
- D) marks <= 40 and attendance <= 75